Sandoval County Groundwater Data: Uses for Our Research

Dataset brief for plotting and project-report analysis | USGS groundwater statistics CSV

Dataset overview

The attached USGS dataset contains 9,462 statistical records for nine groundwater monitoring wells at four named sites (Sandia Site, Lincoln Middle School, Bernalillo Site, and Dome Road Well). Records span approximately 1998–2026 and include monthly, calendar-year, and water-year summaries. Parameter 72019 reports depth to groundwater below land surface (feet); the CSV includes dates, well IDs, coordinates, and summary statistics. These sites do not include Cuba, so findings describe monitored parts of Sandoval County—not Cuba's municipal aquifer.

How the data supports our research questions

Q1 · Groundwater trends — Strong at monitored wells; limited countywide. Plot annual mean depth to water against year by well, then estimate trends (ft/year). An increasing depth means groundwater levels are falling; decreasing depth means they are rising.

Q2 · Well characteristics — Partial. Compare depth to water among wells and over time. The file does not establish total well depth, screened interval, or aquifer formation; these require well logs and geologic records.

Q8 · Monitoring coverage / gaps — Strong for this download. Map the nine well coordinates and summarize each well's first/last observation and data gaps. Eight wells are clustered at three sites; the dataset contains no Cuba wells. This is a gap in this extract, not proof that Cuba has no monitoring.

Q9 · Water-stress dashboard — Strong as a component. Show trend direction/rate, recent versus historical depth, measurement coverage, and a site map. Label results as groundwater-level indicators, not direct measures of aquifer depletion or remaining water supply.

Suggested plots and handling notes

Prioritize (1) annual mean depth-to-water time series by monitoring_location_id; (2) a map of x/y well coordinates; (3) a bar chart of trend slopes (ft/year); and (4) a monitoring-coverage timeline. Filter interval_type = calendar_year for annual trend plots; do not treat monthly, calendar-year, and water-year summaries as independent observations. Check arithmetic_mean_sample_count and missing years before comparing slopes. Keep well IDs distinct even when wells share site names or coordinates.

Interpretation and source

This file is useful for Q1, Q2, Q8, and Q9, especially as evidence of local trends and data limitations. It cannot establish whether Cuba's source aquifer is nearing depletion; that requires Cuba-area well records, municipal withdrawals, and aquifer-specific information. Source: USGS Water Data statistics download (Sandoval County, NM; groundwater; parameter 72019), provided with this brief.

Source:
https://api.waterdata.usgs.gov/download/statistics/?statistics=intervals&site_type_code=GW&computation_type=arithmetic_mean&computation_type=maximum&computation_type=median&computation_type=minimum&computation_type=percentile&state_code=US%3A35&county_code=US%3A35%3A043&normal_type=DOY&normal_type=MOY&interval_type=M&interval_type=CY&interval_type=WY